Department of Computer Science and Engineering
National Institute of Technology Patna
Mid Semester Examination – Odd Sem, October 2023
B. Tech CSE/CE/ECE/EE 5th Sem

F.M: 30
Time: 2hrs

Subject: Machine Learning
Code: CS54118/CS5439

Answer ALL questions. Calculator is allowed. This exam has two pages.

1. Train a Perceptron model to learn the following truth table:

X	Y	Output (Target)
1	1	1
1	0	1
0	1	0
0	0	1

Take the initial parameters as $w_o = 0.01$, $w_1 = -0.03$, $w_2 = -0.04$, $\eta = 0.4$ and bias as -1. Use threshold activation function ($g(h) = 1$, if $h > 0$; and $g(h) = 0$ if $h \leq 0$) and sequential update strategy. Consider input samples in the same order as given in the above table. Show iteration steps. While doing computations you may maintain accuracy up to two places of decimal only for convenience. (**Hint:** Do not work beyond three epochs in your calculations.) **[10 marks]**

2. Consider the following summary of whether golf was played or not (depending upon weather conditions) for the past few days:

Outlook	Temperature	Humidity	Wind	Play
Sunny	85	85	FALSE	Yes
Sunny	80	90	TRUE	Yes
Overcast	83	78	FALSE	No
Rain	70	96	FALSE	No
Rain	68	80	FALSE	No
Rain	65	70	TRUE	Yes
Overcast	64	65	TRUE	No
Sunny	72	95	FALSE	Yes
Sunny	69	70	FALSE	No
Rain	75	80	FALSE	No

Given that it is raining today, temperature is 80°F, humidity is 66%, and wind is blowing, use Naïve Bayes classification to determine if golf will be played today. For simplification, treat:
Temperature ≥ 80 as hot, in range [70,79] as pleasant, and < 70 as cold.
Humidity ≥ 85 as humid, in range [71,84] as average, and < 71 as dry.
Specifically, you need to do the following computations:

- p("No-Play") and p("Yes-Play")

- p(No-Play|Input)=p(Outlook=Rain|No-Play)*p(Temperature=Hot|No-Play)*p(Humidity=Dry|No-Play)*p(Wind=TRUE|No-Play)*p(No-Play)
- p(Yes-Play|Input)=p(Outlook=Rain|Yes-Play)*p(Temperature=Hot|Yes-Play)*p(Humidity=Dry|Yes-Play)*p(Wind=TRUE|Yes-Play)*p(Yes-Play)

Using the above computations, you need to determine whether golf will be played or not. **[6 marks]**

3. (a) Consider a two-class (positive and negative) classification problem. There are 130 samples in my test dataset, out of which 70 are positive. My ML algorithm predicts 80 samples as positive, out of which 50 predictions are correct. Draw the confusion matrix and compute precision, recall and F1 score. **[4 marks]**

(b) Consider the confusion matrix for the following three class classification problem for email classification as "urgent", "normal", and "spam"

Actual Labels \ Predicted Labels	urgent	normal	spam
urgent	8	10	1
normal	5	60	50
spam	3	30	200

Compute the precision and recall values for each of the three classes. **[3 marks]**

(c) Why either of precision or recall alone is not a sufficient measure of accuracy. Explain with real life example(s). **[2 marks]**

4. (a) What is overfitting. Explain how to prevent overfitting during training. **[3 marks]**

(b) What is bias input and what is its need? **[2 marks]**

For Information purpose only:

	CO1	CO2	CO5	CO8	CO10	CO11	Bloom's Level
Q1					OK	OK	2,3
Q2		OK		OK			2,3,4
Q3	OK		OK				2,3
Q4	OK				OK		2,3,4